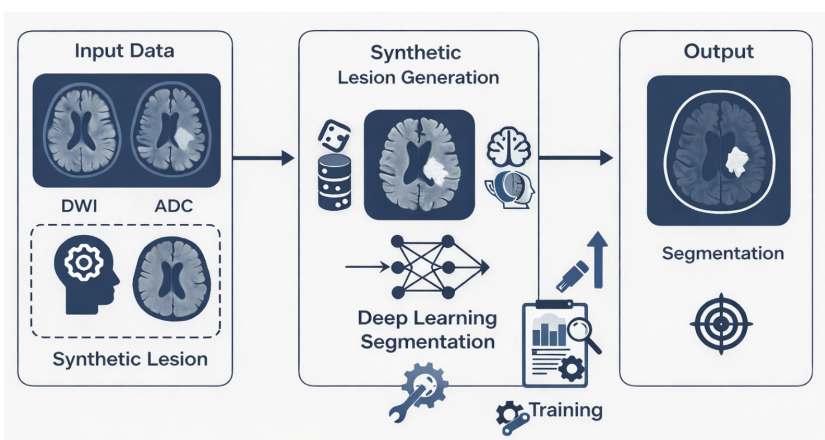
Signal Processing Laboratory 5

January 27, 2026

Semester Project | SMT

Synthetic Stroke Lesion Simulation and Deep Learning Segmentation Using HCP Data



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Abstract

Automatic segmentation of acute ischemic stroke lesions in diffusion MRI (DWI/ADC) is clinically valuable but typically requires large expert-annotated datasets. This project investigates a simulation-driven approach to generate realistic synthetic lesions and to train a deep learning segmentation model using these synthetic labels.

First, we perform a statistical analysis of real lesions using ISLES-2022 to quantify distributions of lesion volume, lesion multiplicity, anatomical localization, intensity contrast in ADC/DWI, and 3D morphological shape descriptors. These measurements define empirical priors for simulation. Second, we implement a lesion generation pipeline that inserts synthetic lesions into healthy high-quality data (HCP) while enforcing multi-modality consistency (ADC decrease, DWI increase), plausible anatomical placement, intensity harmonization, and realistic shape variability. Third, we train a 3D Attention U-Net on mixtures of real ISLES and synthetic datasets and evaluate generalization. Results show consistently strong Dice scores on real ISLES validation data (all configurations > 0.81 Dice), supporting the core hypothesis that simulation-based data can effectively support stroke lesion segmentation. We discuss why simple spherical lesions can yield high Dice, how non-spherical lesions can introduce a domain gap, and how anatomical localization improves precision and loss stability. Finally, we outline future directions to increase biophysical realism (ADC-driven lesion modeling, core/rim contrast, region-specific priors) and to test generalization on completely unseen clinical cohorts.

1 Introduction

Ischemic stroke is a major cause of mortality and long-term disability. In clinical routine, diffusion-weighted imaging (DWI) and apparent diffusion coefficient (ADC) maps are standard for identifying acute infarcts because they are sensitive to early microstructural changes. Acute ischemic lesions typically appear hyperintense on DWI and hypointense on ADC due to restricted diffusion associated with cytotoxic edema. Quantifying lesion location and extent is important for assessing tissue damage and supporting prognosis.

Manual delineation of lesions is time-consuming and observer-dependent. Deep learning has shown strong performance for medical segmentation, but stroke lesion segmentation remains limited by the cost and scarcity of voxel-wise expert annotations and by the heterogeneity of lesion appearance and acquisition protocols. Public datasets such as ISLES provide strong benchmarks but are still limited in scale and diversity [1].

Concretely, this project investigates whether simulation-driven lesion generation can effectively support deep learning–based stroke lesion segmentation when real annotated data are limited. The work is structured in three main stages. First, we perform a detailed statistical analysis of real ischemic lesions from ISLES-2022. These measurements define data-driven priors that characterize how real stroke lesions appear and vary in clinical practice.

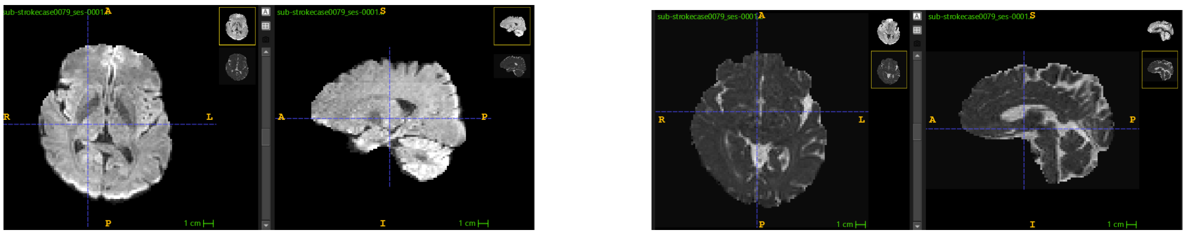
Second, we design and implement a synthetic lesion generation pipeline that inserts artificial ischemic lesions into healthy diffusion MRI scans from the Human Connectome Project. This stage aims to control lesion realism along statistical, anatomical, and geometric dimensions while maintaining exact voxel-level ground-truth masks.

Third, we train a three-dimensional Attention U-Net on mixtures of real ISLES data and synthetic datasets generated under different simulation configurations. The final objective of the project is to determine whether synthetic lesions can act as an effective inductive bias for stroke lesion segmentation, and to identify which aspects of simulation realism are necessary, sufficient, or potentially detrimental for generalization to real ischemic stroke images.

2 Datasets

2.1 ISLES-2022

The **ISLES-2022** dataset, released for the MICCAI 2022 ISLES challenge, provides multi-center diffusion MRI (DWI and ADC) and expert lesion masks for acute ischemic stroke [1]. In this project, ISLES-2022 is used as (i) a source of real lesion statistics that define priors for simulation and (ii) a real clinical benchmark for evaluating model generalization.



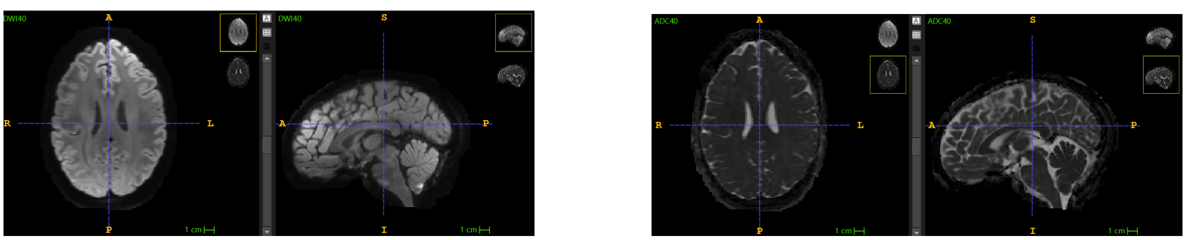
(a) DWI sample from the ISLES dataset

(b) ADC sample from the ISLES dataset

Figure 1: Representative slices from the ISLES-2022 dataset showing (a) Diffusion-Weighted Imaging (DWI) and (b) Apparent Diffusion Coefficient (ADC) maps from acute ischemic stroke patients.

2.2 Human Connectome Project

The **Human Connectome Project (HCP)** provides high-quality MRI acquired under standardized protocols on healthy subjects [2]. In our pipeline, HCP scans are used as lesion-free anatomical substrates on which synthetic lesions are inserted. This choice provides clean background anatomy, consistent acquisition, and strong image quality, enabling controlled simulation while avoiding patient privacy concerns.



(a) DWI sample from the HCP dataset

(b) ADC sample from the HCP dataset

Figure 2: Representative slices from the Human Connectome Project (HCP) dataset showing (a) Diffusion-Weighted Imaging (DWI) and (b) Apparent Diffusion Coefficient (ADC) maps from healthy control subjects.

3 Statistical Analysis for Lesion Generation

This part quantifies the empirical properties of real ischemic lesions to constrain simulation. The objective is not only to reproduce the mean appearance of lesions, but to reproduce the variability that makes stroke segmentation challenging.

3.1 Preprocessing rationale and notation

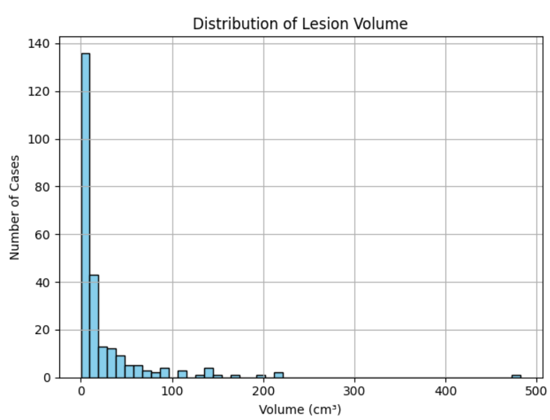
All NIfTI volumes and associated spatial metadata (voxel spacing, affine transforms) were handled using the NiBabel library [3].

To measure statistics consistently across subjects, volumes must be comparable in space and intensity. Preprocessing therefore aims to reduce confounding factors such as voxel size differences, orientation differences, and scanner-dependent intensity scaling. Lesion masks are analyzed in 3D using connected components (26-connectivity, i.e., faces/edges/corners). Very small components are excluded from specific analyses to reduce sensitivity to spurious isolated detections (typical thresholds used in this project: < 5 voxels for lesion-count studies; < 10 voxels for more stable shape descriptor estimation).

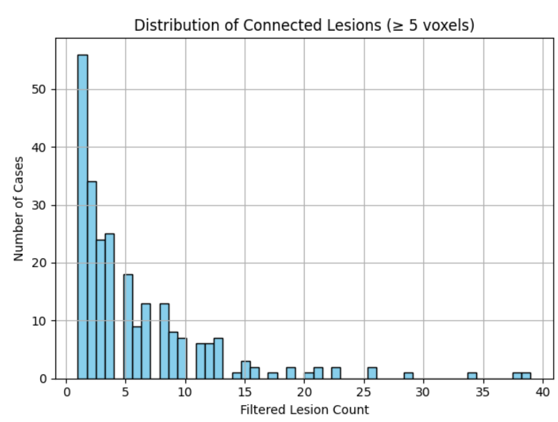
We denote by $M(\mathbf{x}) \in \{0, 1\}$ the binary lesion mask, by $\Omega = \{\mathbf{x} : M(\mathbf{x}) = 1\}$ the lesion voxel set, and by $V = |\Omega|$ the lesion volume in voxels. When converting to physical volume, we use $V_{\mathrm{mm}^3} = V \cdot s_x s_y s_z$ where (s_x, s_y, s_z) are voxel spacings in mm.

3.2 Lesion volume and lesion multiplicity

Lesion volume distributions exhibit a long tail: many subjects have small-to-moderate infarcts, while a minority show very large infarct volumes (Fig. 3a). Lesion multiplicity (number of connected components) varies substantially and can be high even when total lesion volume is not extreme (Fig. 3b). This distinction matters because a model must detect both large territorial lesions and multiple scattered embolic lesions; simulation should therefore represent both patterns.
(Code: `strokevolume.py`, `statisticalanalysis.py`)



(a) Distribution of total lesion volumes across ISLES-2022 subjects.

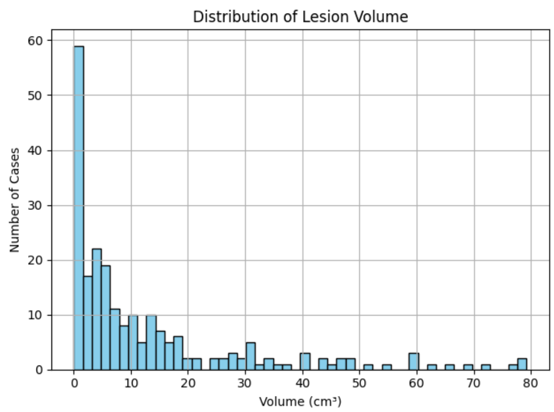


(b) Distribution of the number of connected lesion components across ISLES-2022 subjects (26-connectivity).

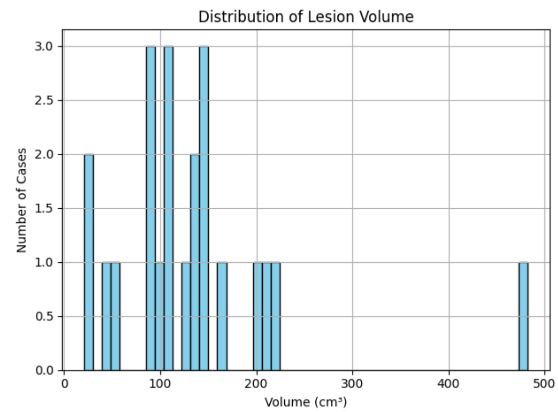
Figure 3: Statistical distributions of lesion characteristics from the ISLES-2022 dataset: (a) lesion volume distribution showing the total infarct volume per subject, and (b) distribution of connected lesion components per subject, highlighting the multiplicity of ischemic foci.

3.3 Inlier vs. outlier analysis

Outliers are clinically relevant because rare lesion patterns still appear in practice and can break models that only learn "typical" cases. We detect outliers using Tukey's interquartile range (IQR) with a conservative multiplier of 3, which isolates cases with unusually large volume and/or unusually high lesion counts. In the analyzed cohort, approximately 8% (22 cases) fall into this category. In later simulation, this motivates either controlled sampling of extreme cases or explicit stress testing of models. Extensive details about the implemntation of this part of the analysis can be found in `statisticalanalysis.py`



(a) Distribution of lesion volumes for inlier subjects.



(b) Distribution of lesion volumes for outlier subjects.

Figure 4: Comparison of lesion volume distributions between inlier and outlier subjects in the ISLES-2022 dataset. (a) Inlier subjects show a typical distribution with most lesions concentrated in smaller volume ranges. (b) Outlier subjects, identified using Tukey's IQR method (multiplier=3), exhibit extreme lesion volumes that deviate significantly from the population norm.

3.4 Intensity contrast in DWI and ADC

A synthetic lesion must be realistic in its local contrast. Instead of relying on global intensity values, we estimate lesion-to-background contrast by comparing intensities inside lesion masks to intensities in the peri-lesional neighborhood. The neighborhood is obtained by a morphological dilation of the lesion mask. This approach reflects how lesions are interpreted clinically: lesions are perceived relative to surrounding tissue. More details can be found in `statisticalanalysis.py`

The observed intensity behavior matches expected acute ischemia: lesions are hyperintense on DWI and hypointense on ADC (Figs. 5–6). These distributions define sampling ranges for synthetic lesion contrast.

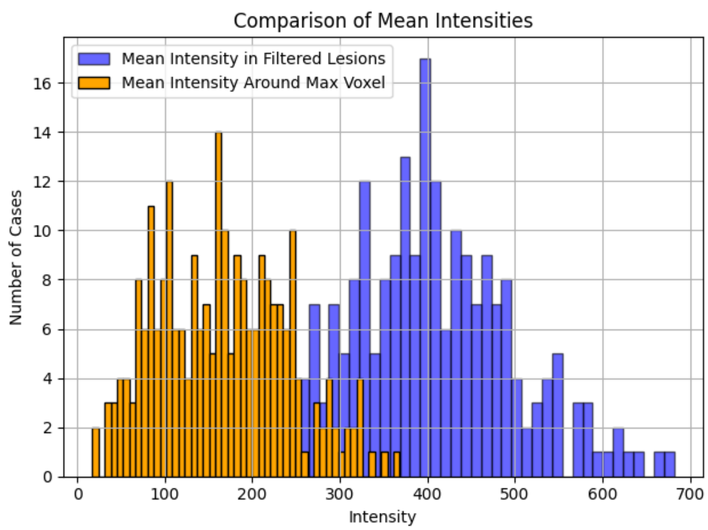


Figure 5: Distribution of mean DWI intensities within lesion masks across subjects.

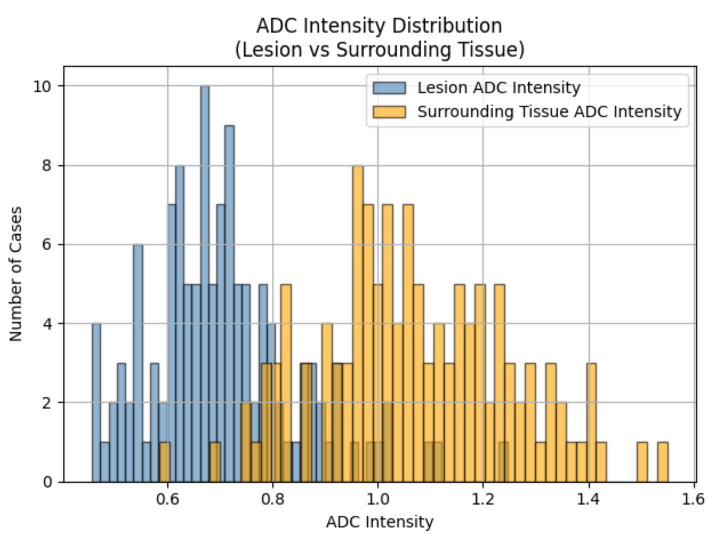


Figure 6: Distribution of mean ADC intensities within lesion masks across subjects.

3.5 Conditional dependencies between lesion count and volume

To capture dependencies between lesion multiplicity and lesion burden, conditional probabilities are estimated, for instance

$$P\big(V_{\max} > v_t \mid N_{\text{lesions}}\big), \qquad P\big(V_{\text{total}} \mid N_{\text{lesions}}\big). \tag{1}$$

The resulting trends suggest that patients with few large lesions are more common than those with many large lesions, and that total lesion volume increases sublinearly with lesion count. This motivates simulation schemes where lesion count and lesion sizes are not sampled independently.

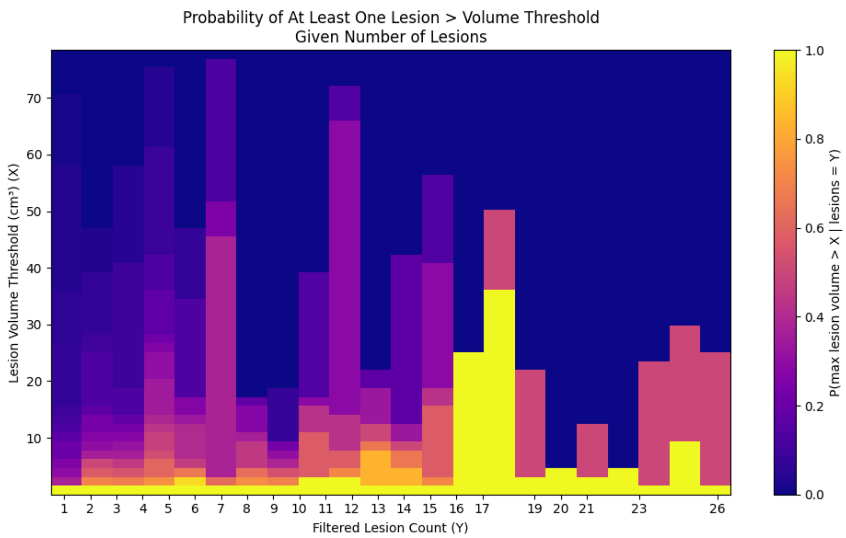


Figure 7: Conditional probability heat map relating lesion count to the probability of large lesions (inliers).

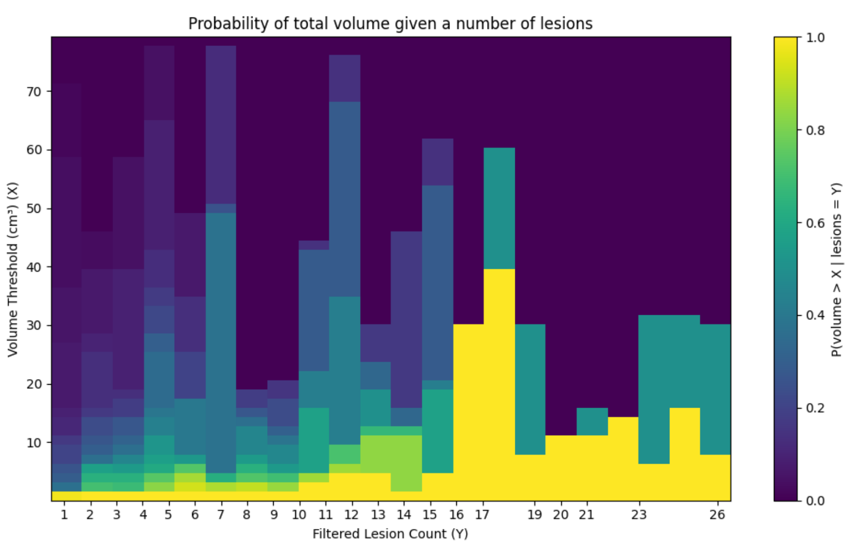


Figure 8: Conditional probability heat map of total lesion volume as a function of lesion count (inliers).

3.6 Anatomical localization using SynthSeg

Lesions are not uniformly distributed across brain tissue. To quantify plausible lesion locations, we use SynthSeg anatomical parcellations [6] and assign each lesion to the region containing its center of mass (code : `localization.py`).

SynthSeg is a learning-based brain MRI segmentation framework designed to produce robust anatomical parcellations across a wide range of contrasts, resolutions, and acquisition protocols without requiring retraining [6]. It relies on a convolutional neural network trained exclusively on synthetically generated brain images, where anatomical labels are known by construction. During training, synthetic images are generated by sampling anatomical label maps, simulating MRI acquisition effects (contrast variation, resolution changes, intensity non-linearities, noise, and bias fields), and then training the network to recover the original labels.

Aspect	**Typical values / ranges**
Model architecture	3D U-Net–like convolutional neural network
Training data	Fully synthetic MRI volumes with known anatomical labels
Contrast simulation	Random linear and nonlinear intensity mappings
Noise modeling	Additive Gaussian noise, $\sigma = 0.1$ (relative units)
Bias field simulation	Low-frequency bias fields, up to $\pm 30\%$ intensity variation
Resolution augmentation	Voxel sizes sampled at 2.5 mm (isotropic)
Spatial transformations	Random rotations, scaling, and shearing
Output labels	31 brain regions (cortex, WM, GM, CSF, etc.)
Inference configuration	Pretrained model, no fine-tuning

Table 1: Key SynthSeg training assumptions and numerical parameter ranges (standard pretrained model).

The observed distributions show a predominance of lesions in cerebral cortex, followed by cerebellar cortex and cerebral white matter. These statistics motivate a location prior used later during synthetic placement.

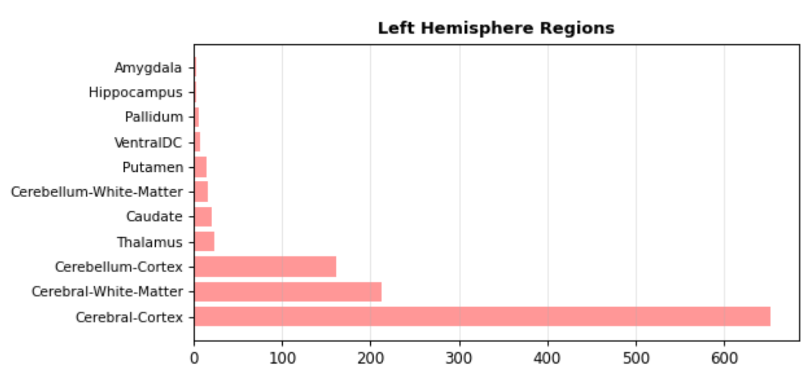


Figure 9: Distribution of lesion locations in the left hemisphere based on anatomical parcellation.

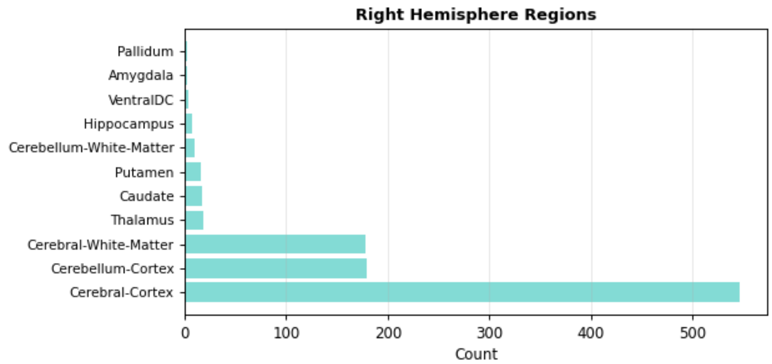


Figure 10: Distribution of lesion locations in the right hemisphere based on anatomical parcellation.

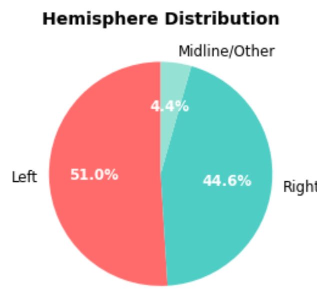


Figure 11: Overall hemispheric distribution of lesion centers.

3.7 Spatial lesion-center prior via kernel density estimation

Beyond region counts, we build a continuous spatial prior for lesion centers. For each connected lesion, we compute its center of mass $\mathbf{c} = (c_x, c_y, c_z)$ in voxel coordinates, and normalize it by image dimensions $\mathbf{d} = (D_x, D_y, D_z)$:

$$\tilde{\mathbf{c}} = \left(\frac{c_x}{D_x}, \frac{c_y}{D_y}, \frac{c_z}{D_z} \right) \in [0, 1]^3. \tag{2}$$

We then fit a 3D Gaussian kernel density estimate (KDE) $p(\tilde{\mathbf{c}})$ with a fixed bandwidth (in our implementation, a scalar bandwidth parameter of 0.1). Sampling from this KDE during simulation biases lesion placement towards clinically frequent locations, while still producing diversity

3.8 Morphological descriptors and shape priors

Lesion shape is a major source of segmentation difficulty. We compute 3D descriptors for each lesion and estimate empirical distributions that define a morphological prior. In voxel space, the implementation uses (epsilons are used to avoid a zero-division):

- **Surface proxy:** surface voxels are estimated by $S = \Omega \, \Delta \, \mathrm{erode}(\Omega)$ (XOR between mask and its binary erosion), and $A = |S|$ is used as a proxy for surface area.
- **Convex hull:** a convex hull is computed from voxel coordinates, yielding a hull volume V_{hull} and hull surface area A_{hull}.
- **Eccentricity:** computed from eigenvalues of the covariance of lesion voxel coordinates,

  $$e = \sqrt{1 - \frac{\lambda_{\min}}{\lambda_{\max}}}\,, \tag{3}$$

  where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest eigenvalues.
- **Convexity:** $C_{\mathrm{vx}} = V/(V_{\mathrm{hull}} + \varepsilon)$.
- **Roughness:** $R = A/(A_{\mathrm{hull}} + \varepsilon)$, i.e. how much extra surface is present compared to a convex reference.
- **Compactness:** a dimensionless compactness proxy

  $$K = \frac{V^2}{A^3 + \varepsilon}\,, \tag{4}$$

  which is equivalent (up to a cubic transform) to the common $V^{2/3}/A$ compactness used in shape generation.
- **Roundness (sphericity):**

  $$\Phi = \frac{\pi^{1/3}(6V)^{2/3}}{A + \varepsilon} \in (0, 1]\,, \tag{5}$$

  which equals 1 for an ideal sphere in continuous geometry (up to voxelization effects).

These priors enable us to generate non-spherical lesions whose geometry matches clinical variability rather than relying only on simple spheres. Figure 12 illustrates empirical distributions. (Code: `morphologicalanalysis.py`)

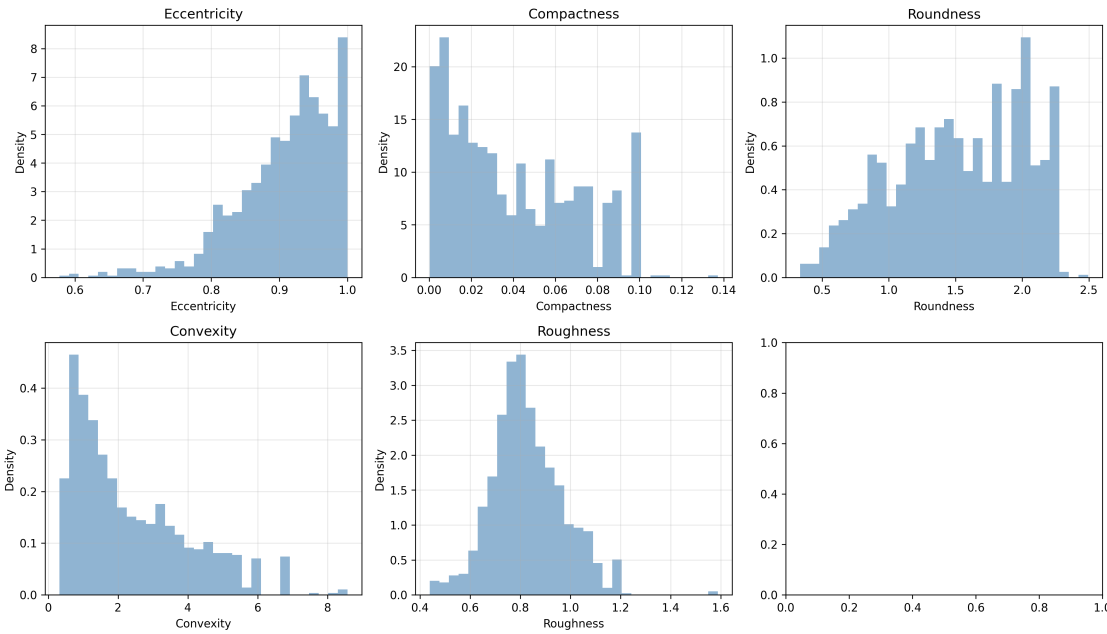


Figure 12: Empirical distributions of morphological shape descriptors estimated from ISLES lesions.

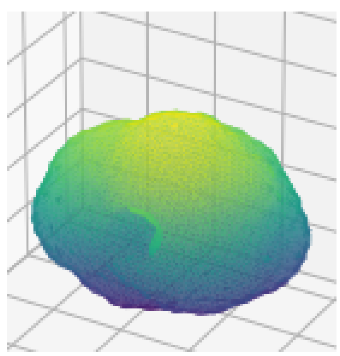


Figure 13: Example output of the morphology-driven synthetic lesion shape generator.

4 Synthetic Lesion Generation

The synthetic lesion generator is designed to produce paired (ADC, DWI) volumes and an exact binary lesion mask, while remaining statistically and anatomically consistent with clinical data. The goal is not photorealism alone, but *training usefulness*: the synthetic lesions should provide inductive biases that help models generalize to real clinical stroke.

For the implementation of the generator, please refer to `shapegenerationV2.py`

4.1 Overview and notation

Let $I_{\mathrm{ADC}}(\mathbf{x})$ and $I_{\mathrm{DWI}}(\mathbf{x})$ be the healthy substrate images (from HCP) after preprocessing/harmonization. The generator outputs modified images $\hat{I}_{\mathrm{ADC}}, \hat{I}_{\mathrm{DWI}}$ and a lesion mask M. Lesions are produced as a union of N connected components:

$$M = \bigvee_{i=1}^{N} M_i, \qquad M_i \cap M_j = \emptyset \text{ for } i \neq j, \tag{6}$$

where non-overlap is enforced explicitly to avoid implausible merging of independent lesions.

4.2 Why intensity harmonization is needed

Even when two datasets contain the same modality names (ADC, DWI), intensity scales can differ due to acquisition settings, reconstruction, preprocessing, and unit conventions (ADC stored in m^2/s vs. mm^2/s). If synthetic lesions are inserted into HCP using an HCP-native intensity regime, a model may learn features that do not transfer to clinical ISLES intensities. We therefore apply two harmonization steps whose scientific purpose is to reduce domain shift so that "lesion contrast" is comparable between synthetic and real domains. This is done in `intensityshift.py`

4.2.1 ADC unit normalization

A practical issue is that ADC can be stored in m^2/s (typical values $\sim 10^{-6}$) or mm^2/s (typical values $\sim 10^{-3}$). The implemented heuristic checks the median ADC intensity within brain voxels and applies a factor 10^6 when values are consistent with m^2/s, thereby converting to mm^2/s. This prevents generating lesions with unrealistic absolute ADC contrast when mixing datasets.

4.2.2 Histogram matching via percentile mapping

After unit standardization, histogram matching maps the source intensity distribution to a reference distribution built from ISLES. Using percentile-based mapping, we estimate empirical cumulative distributions using P percentiles (here $P = 1000$) and apply:

$$\hat{I}(\mathbf{x}) = F_{\mathrm{ref}}^{-1}\big(F_{\mathrm{src}}(I(\mathbf{x}))\big), \tag{7}$$

implemented numerically by interpolating between percentile landmarks. ADC and DWI are matched separately to preserve their modality-specific contrast while reducing global scaling differences between HCP and ISLES.

4.3 Anatomical constraints and tissue validity masks

Lesions should not be placed in non-brain voxels or ventricular CSF. We therefore create an *allowed tissue mask* $A(\mathbf{x}) \in \{0, 1\}$ derived from SynthSeg labels:

$$A(\mathbf{x}) = \mathbb{1}[\text{label}(\mathbf{x}) > 0] \cdot \mathbb{1}[\text{label}(\mathbf{x}) \notin \mathcal{V}], \tag{8}$$

where $\mathcal{V}$ is a set of ventricular/CSF-related labels excluded from placement (e.g. lateral ventricles, 3rd/4th ventricle). SynthSeg maps are resampled to the diffusion grid using nearest-neighbor interpolation to preserve discrete labels. This constraint substantially increases anatomical plausibility and reduces "shortcut" learning on synthetic artifacts.

4.4 Sampling lesion count and target volume

To reflect real stroke variability, lesion count and lesion size can be either randomly sampled in controlled ranges or sampled from empirical distributions inferred from ISLES.

Lesion count. When distribution control is disabled, N is sampled uniformly in a small range (e.g. $N \in \{1, \ldots, 5\}$). When enabled, N is sampled from an empirical discrete distribution derived from ISLES connected-component counts (after filtering small components).

Target volume in voxels. For each lesion, a target volume may be sampled in cm^3 based on ISLES statistics and converted to voxels using voxel spacing:

$$V_{\text{vox}} = \left\lceil \frac{V_{cm^3}}{(s_x s_y s_z) \cdot 10^{-3}} \right\rceil, \tag{9}$$

since $1\,mm^3 = 10^{-3}\,cm^3$. This conversion ensures that "volume priors" remain comparable even when voxel spacing differs across datasets.

4.5 Lesion center sampling: random vs. KDE-guided placement

Lesion centers can be sampled uniformly within $A(\mathbf{x}) = 1$ or from the learned KDE prior described in Section 3.7. Using KDE-guided sampling, we draw $\tilde{\mathbf{c}} \sim p(\tilde{\mathbf{c}})$ and map it to voxel coordinates $\mathbf{c} = \lfloor \tilde{\mathbf{c}} \odot \mathbf{d} \rfloor$. If $\mathbf{c}$ falls outside valid tissue, the implementation attempts a local correction by selecting the nearest allowed voxel within a bounded radius (e.g. 20 voxels). If this fails, it resamples or falls back to uniform placement.

Scientifically, KDE-guided placement serves as an inductive bias: it increases the frequency of lesions in vascularly plausible regions and reduces the number of anatomically implausible training examples.

4.6 Shape generation: spherical lesions (minimal setting)

Spherical lesions are used as the minimal simulation setting. They allow us to answer the feasibility question: *is a simple synthetic lesion model already sufficient to improve or maintain segmentation performance on real clinical data?*

A key implementation detail is that "spheres" are grown inside the connected component of the allowed mask that contains the seed voxel. This prevents a sphere from leaking into invalid tissue (e.g. crossing ventricles). Concretely:

1. Choose seed voxel $\mathbf{c}$ inside valid tissue.

2. Extract the connected component C of $A(\mathbf{x}) = 1$ that contains $\mathbf{c}$.

3. Compute physical distance $d(\mathbf{x})$ from $\mathbf{c}$ for all $\mathbf{x} \in C$:

$$d(\mathbf{x}) = \sqrt{\big((x - c_x)s_x\big)^2 + \big((y - c_y)s_y\big)^2 + \big((z - c_z)s_z\big)^2}. \tag{10}$$

4. Select the V_{vox} voxels with smallest distance to form M_i.

This produces a smooth, compact lesion constrained by anatomy and with an exact voxel count.

4.7 Shape generation: non-spherical lesions via a shape library

To better approximate irregular lesion boundaries, we also generate lesions from non-spherical templates. In this project stage, 16 distinct template shapes were used.

4.7.1 Morphology-driven template generation (optimization view)

Template shapes are generated offline using distributions of morphological descriptors from ISLES. We sample target descriptor values (e.g. eccentricity, convexity, compactness, roundness) from KDEs and a target volume from a Gamma prior. Starting from an initial sphere, the generator optimizes a parametric 3D mask (ellipsoid parameters + surface noise) using a multi-objective loss:

$$\mathcal{L}(\theta) = \sum_{k \in \mathcal{K}} w_k \left(\frac{m_k(\theta) - t_k}{t_k + \varepsilon} \right)^2, \tag{11}$$

where $m_k(\theta)$ is the measured descriptor for parameters θ, t_k is the sampled target, and w_k are weights. Optimization uses derivative-free methods, such as the Nelder–Mead simplex method [4] and Powell's conjugate direction method [5].. When optimization fails, a fallback procedure performs gentle eccentricity adjustments and volume rescaling to approach the target while maintaining plausibility.

4.7.2 Embedding and scaling of template shapes

Given a binary template T with volume V_T, we rescale it to a desired target volume V_{vox} using an isotropic scaling factor

$$s = \left(\frac{V_{\text{vox}}}{V_T} \right)^{1/3}. \tag{12}$$

The rescaled template is then placed so that its centroid aligns with the chosen lesion center. After placement, it is intersected with the allowed tissue mask and with the complement of previously occupied voxels. If the final mask exceeds the target voxel count, random subsampling is applied to match the exact volume; if too few voxels remain due to anatomical clipping, a resample of center/shape is performed.

This step introduces a realistic boundary complexity but can also create a *domain gap* if the template boundary statistics (e.g. roughness) differ from those in real ISLES lesions.

4.8 Multi-modality intensity model (ADC decrease, DWI increase)

A lesion mask is inserted consistently into both modalities, enforcing complementary signal changes. The current intensity model is intentionally simple but controlled:

$$\hat{I}_{\mathrm{ADC}}(\mathbf{x}) = I_{\mathrm{ADC}}(\mathbf{x}) \cdot \big(1 - M(\mathbf{x})\big) + \alpha_{\mathrm{ADC}}\, I_{\mathrm{ADC}}(\mathbf{x}) \cdot M(\mathbf{x}), \tag{13}$$

$$\hat{I}_{\mathrm{DWI}}(\mathbf{x}) = I_{\mathrm{DWI}}(\mathbf{x}) \cdot \big(1 - M(\mathbf{x})\big) + \alpha_{\mathrm{DWI}}\, I_{\mathrm{DWI}}(\mathbf{x}) \cdot M(\mathbf{x}), \tag{14}$$

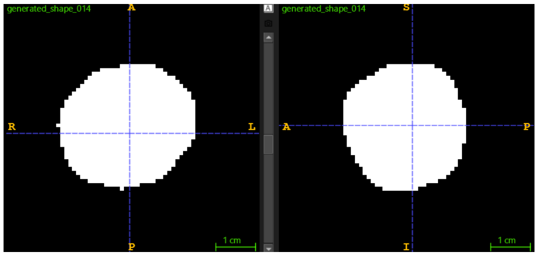
with $\alpha_{\mathrm{ADC}} < 1$ and $\alpha_{\mathrm{DWI}} > 1$. In the implemented default ranges, $\alpha_{\mathrm{ADC}} \sim U(0.5, 0.9)$ and $\alpha_{\mathrm{DWI}} \sim U(1.7, 2.2)$, while optionally sampling these factors from ISLES-derived distributions.

To reduce boundary sharpness and emulate peri-lesional effects, we also define a surrounding region

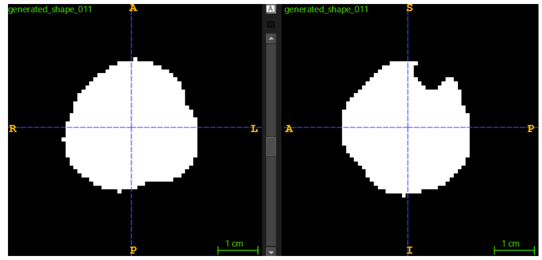
$$R = \mathrm{dilate}(M, r) \setminus M, \tag{15}$$

using a dilation radius of $r = 5\,\mathrm{mm}$ (converted to voxels using spacing). A mild multiplicative factor α_{sur} can be applied within R to create a transition zone (less intensity surrounding the lesion, to be biologically accurate).

4.9 Qualitative examples

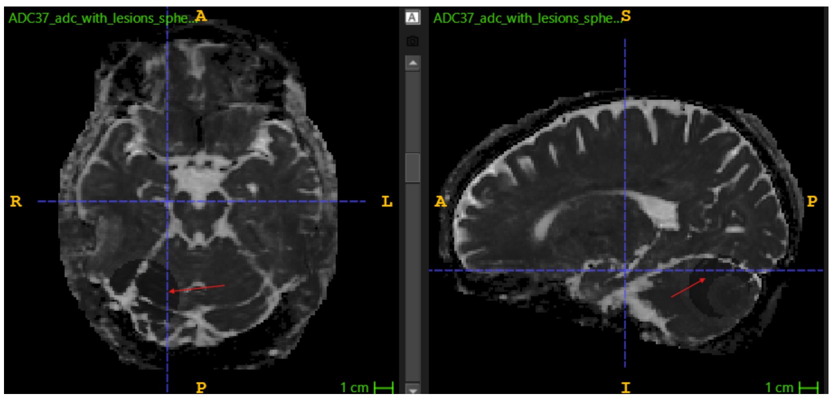


(a) Template shape example 1

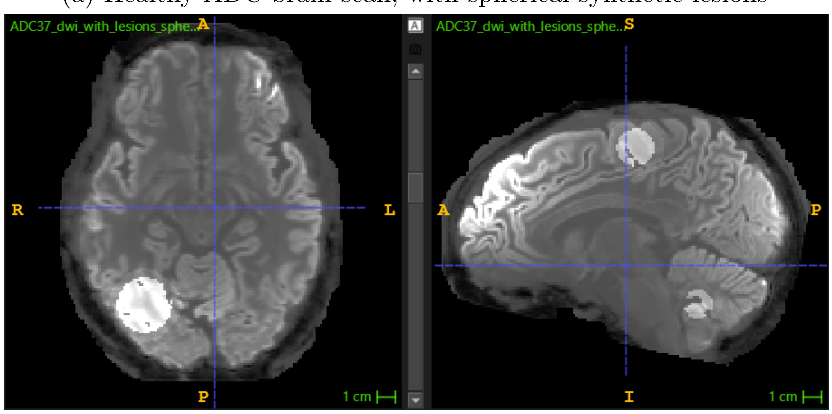


(b) Template shape example 2

Figure 14: Examples of non-spherical lesion templates from the morphology-driven shape library.

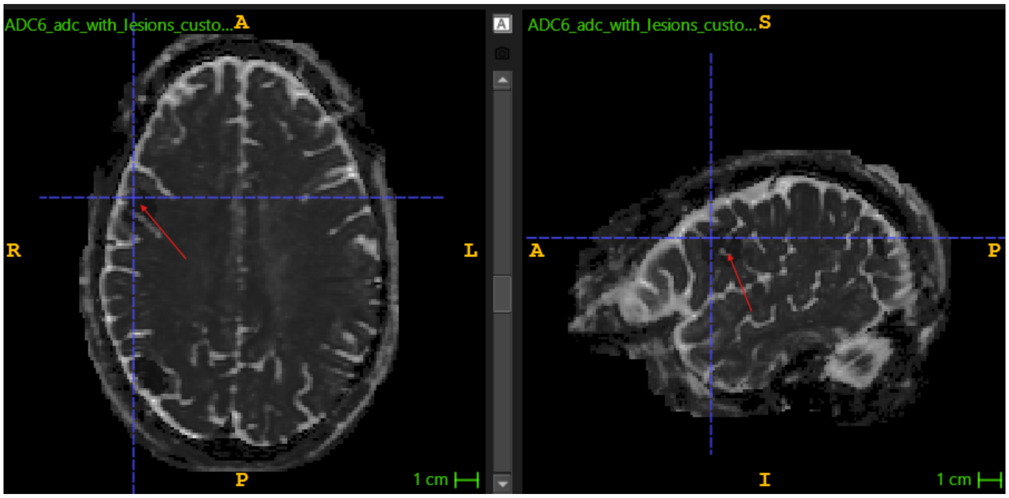


(a) Healthy ADC brain scan, with spherical synthetic lesions

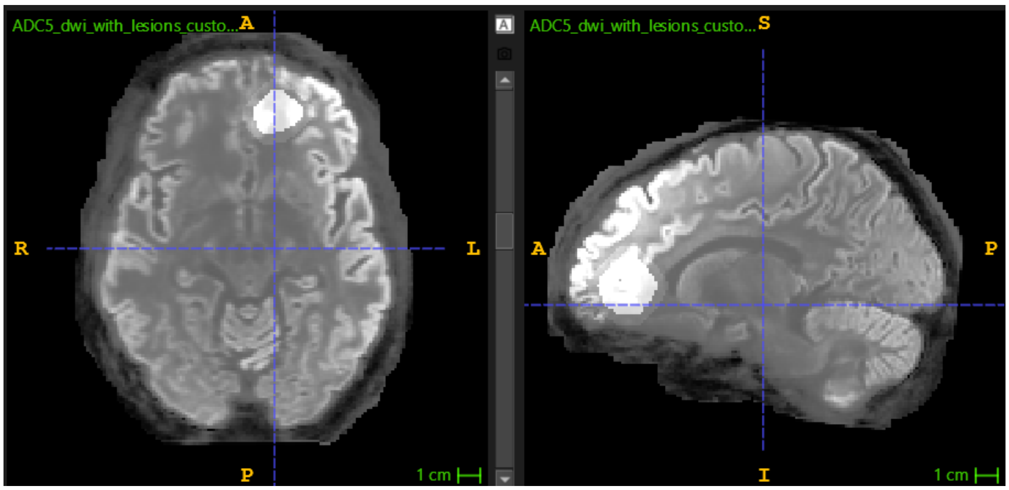


(b) Template shape example 2

Figure 15: Healthy DWI brain scan, with spherical synthetic lesions



(a) Healthy ADC brain scan, with non-spherical synthetic lesions



(b) Healthy DWI brain scan, with non-spherical synthetic lesions

Figure 16: Examples of non-spherical lesion templates from the morphology-driven shape library.

5 Deep Learning Segmentation using Synthetic Lesions

This section evaluates whether synthetic lesions meaningfully support training of a stroke lesion segmentation model and how different simulation choices affect performance.

5.1 Model: what is an Attention U-Net?

U-Net is a widely used encoder–decoder architecture for biomedical segmentation [7]. It progressively downsamples feature maps to capture global context and upsamples them back to full resolution, using skip connections to preserve spatial detail.

Attention U-Net extends U-Net by introducing *attention gates* on skip connections [8]. Instead of directly concatenating encoder features into the decoder, the model learns to weight spatial regions of the skip features based on a gating signal coming from coarser decoder representations. Conceptually, attention suppresses irrelevant background activations and highlights task-relevant regions. This is particularly well suited to stroke lesion segmentation because lesions are often small relative to the brain volume, and naive skip connections can propagate strong background texture that increases false positives.

A common formulation of an attention gate computes attention coefficients $\alpha(\mathbf{x}) \in [0, 1]$ from skip features x and a gating signal g:

$$\alpha = \sigma\Big(\psi^\top \operatorname{ReLU}(W_x x + W_g g + b)\Big), \qquad \tilde{x} = \alpha \odot x, \tag{16}$$

where σ is a sigmoid, $\odot$ denotes element-wise multiplication, and $\tilde{x}$ is the gated skip feature passed to the decoder.

5.2 Implementation used in this project (3D Attention U-Net)

We train a **3D** Attention U-Net implemented in MONAI [9], with two input channels corresponding to concatenated ADC and DWI volumes. The network outputs a single-channel lesion probability map (sigmoid activation at inference). The main architecture parameters are:

- Spatial dimension: 3D volumes (patch-based training).
- Input channels: 2 (ADC and DWI concatenated).
- Encoder channels: $(16, 32, 64, 128)$ with strides $(2, 2, 2)$.
- Dropout: 0.1.
- Training : 85 % Validation : 15 %

The implementation details can be found in `train.py` (from Haoxuan Wang master's project - 2024)

5.3 Training strategy: why these choices are made

The training pipeline is designed around the primary difficulty of lesion segmentation: **extreme class imbalance** (very few lesion voxels) and **high variability** in lesion appearance.

Voxel spacing standardization. Inputs are resampled to a fixed voxel spacing (approximately $(1, 1, 2)$ mm) to reduce variability due to voxel geometry and to make spatial patterns more comparable across subjects.

Patch-based learning. Full 3D volumes are memory-intensive. We therefore train on patches of size $(96, 96, 16)$. This also increases the number of training samples and allows explicit control over sampling lesion-containing patches.

Class-balanced cropping. Patches are sampled using a label-guided strategy. In training, a ratio of approximately $(1 : 2)$ background-to-lesion sampling is used to prevent collapse to trivial background predictions. In validation, a much more background-dominant ratio (e.g. $(1 : 100)$) can be used to better reflect real prevalence while still measuring lesion performance. Multiple patches (here 8) are sampled per volume per epoch to stabilize estimates.

Intensity clamping and scaling. Before normalization, intensities are clamped to robust ranges to reduce sensitivity to outliers (typical values: ADC clamped to $[10, 3000]$, DWI clamped to $[5, 1800]$ in dataset-specific units). Intensities are then scaled to standardized ranges prior to network input. This encourages stable optimization and reduces dependence on scanner-specific scaling.

Augmentation. The training pipeline incorporated data augmentation to improve robustness to anatomical and scanner variability. Two main augmentations were used:

- **Random affine** (prob. 0.3): rotations up to ± 0.1 rad, shearing up to ± 0.05, and scaling up to $\pm 10\%$.
- **Gaussian noise** (prob. 0.5): additive zero-mean Gaussian noise (std. 1.0), applied independently per modality.

These transforms (from the `monai.transforms` library) are applied jointly to images and masks (with appropriate interpolations) to preserve modality alignment.

5.4 Optimization, learning rate schedule, and losses

Optimizer and schedule. Training uses Adam [10] with learning rate 10^{-2}. A cosine annealing learning-rate schedule [11] is used to gradually reduce the learning rate:

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\pi \frac{t}{T_{\max}}\right)\right), \tag{17}$$

with typical values $\eta_{\min} = 10^{-4}$ and $T_{\max} = 200$ epochs. Mixed-precision training (automatic casting + gradient scaling) is enabled when running on CUDA to improve memory efficiency.

Loss function. We use a combined Dice + cross-entropy style loss (MONAI DiceCELoss with sigmoid) to balance overlap-based optimization with voxel-wise calibration:

$$\mathcal{L} = \lambda_{\text{dice}} \, \mathcal{L}_{\text{dice}} + \lambda_{\text{ce}} \, \mathcal{L}_{\text{bce}}, \tag{18}$$

with $\lambda_{\text{dice}} = \lambda_{\text{ce}} = 0.3$ in the current configuration. Dice loss is defined as $\mathcal{L}_{\text{dice}} = 1 - \text{Dice}$ (optionally with smoothing), and BCE is the standard binary cross-entropy on probabilities after sigmoid.

5.5 Evaluation metrics

Let TP, FP, FN denote voxel-wise true positives, false positives, and false negatives after thresholding the predicted probability map at 0.5. We report:

$$\text{Dice} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN} + \varepsilon}, \tag{19}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP} + \varepsilon}, \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN} + \varepsilon}. \tag{21}$$

These metrics reflect complementary aspects: Dice captures overlap, precision penalizes false positives, and recall penalizes missed lesion voxels.

5.6 Experiment design

Four training configurations were evaluated (real ISLES combined with synthetic data variants):

1. **ISLES + spherical synthetic lesions.** Minimal simulation to answer: is this enough?
2. **ISLES + non-spherical synthetic lesions.** Increased boundary realism using 16 shapes.
3. **ISLES + non-spherical lesions + anatomical localization prior.** KDE-guided placement + tissue constraints as inductive bias.
4. **ISLES + non-spherical + anatomical localization + intensity priors.** Add intensity factor sampling/controls to better match clinical contrast distributions.

In addition, a practical sanity-check protocol was used during development: the four same subjects were removed from the dataset at each run for manual inspection. Qualitatively, the trained model generalized well to the synthetic additions, suggesting that the network learned transferable lesion cues rather than memorizing specific cases.

5.7 Quantitative results

Table 2 summarizes key training and validation metrics. "Val Dice (all)" is the main selection criterion reported in logs; "Val Dice (ISLES)" is computed on real ISLES validation data; and "Val Dice (synthetic)" corresponds to the synthetic validation subset.

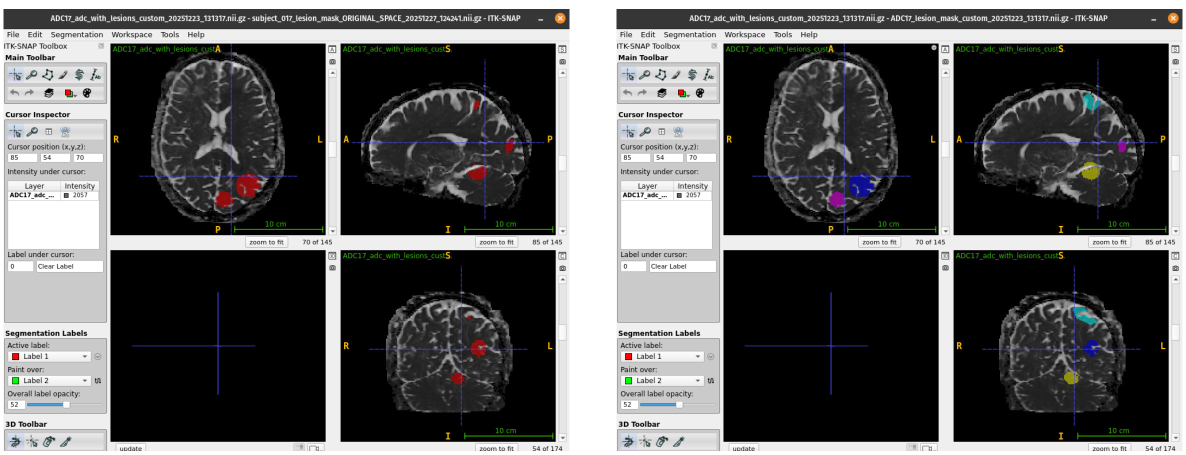
Table 2: Segmentation results for different synthetic training configurations.

Training configuration	Train Dice	Train loss	Val Dice (all)	Val Dice (ISLES)	Val Dice (synthetic)	Val Precision (all)	Val Recall (all)	Val CE (all)	Val Dice loss (all)
Baseline (Attention U-Net, ISLES'22)	–	–	0.726 ±0.004	0.726 ±0.004	–	–	–	–	–
ISLES + spherical synthetic lesions	0.8319	0.3837	**0.8517**	0.8278	0.8329	0.8553	**0.8700**	0.1042	**0.1591**
ISLES + non-spherical synthetic lesions (16 shapes)	**0.8534**	**0.3571**	0.8435	0.8142	**0.8599**	0.8475	0.8625	0.0833	0.1723
ISLES + non-spherical + anatomical localization prior	0.8406	0.3648	0.8422	0.8182	0.8554	0.8651	0.8460	**0.0659**	0.1627
ISLES + non-spherical + localization + intensity priors	0.8420	0.3596	0.8272	**0.8286**	0.8073	**0.8665**	0.8110	0.0801	0.1905

5.8 Limitations of the current evaluation

The reported pipeline is patch-based and uses dataset-specific preprocessing and sampling. A strict comparison to published numbers requires matching evaluation protocols, splits, and full-volume inference settings. In addition, the observation that results appear better than some previously reported baselines is encouraging but should be confirmed on completely unseen external datasets to ensure that performance is not specific to the current evaluation setup.

5.9 Visual inference



(a) ADC image with synthetic ischemic lesions overlaid.

(b) Corresponding synthetic lesion mask (ground truth)

Figure 17: Example of synthetic lesion generation on a healthy ADC scan. The left panel shows the ADC image with inserted synthetic lesions, while the right panel shows the corresponding ground-truth lesion mask used for supervised training.

This inference illustrates a challenging case, in which the model under-segments (top-right, on the sagittal slice) a lesion component. Even though the other lesions are well segmented, some lesions can be harder to segment. These cases are discussed in the next section.

6 Discussion and Future Work

The experimental results reveal several concrete behaviors that clarify both the strengths and limitations of simulation-based lesion training. Across all configurations, segmentation performance on real ISLES validation data remains consistently high (Dice > 0.81; Table 2), indicating that synthetic lesions, when used as augmentation, do not degrade generalization and can effectively support learning despite their artificial origin. However, closer inspection of predictions highlights important qualitative differences between simulation strategies that are not fully captured by Dice alone.

In the spherical-lesion configuration, the model achieves the highest overall validation Dice and strong recall. Visual inspection shows that these models tend to produce compact, well-localized predictions, but occasionally over-segment lesion boundaries, especially for elongated or irregular real infarcts. This behavior is consistent with the regular geometry of spherical training examples: overlap-based metrics favor smooth shapes, and the model learns a strong bias toward compact connected components. As a result, Dice remains high even when boundary accuracy is imperfect, explaining why this minimal simulation setting performs surprisingly well under overlap-based evaluation.

In contrast, models trained with non-spherical synthetic lesions demonstrate improved performance on the synthetic validation subset (higher synthetic Dice in Table 2), but a slight decrease in Dice on real ISLES data which is quite surprising. Qualitatively, these models better capture irregular lesion outlines but also produce more fragmented predictions and occasional false positives in regions with complex anatomy. In particular, small spurious detections were more frequently observed near ventricular borders and deep white matter, where synthetic shape boundaries interact with partial-volume effects. This suggests that increasing geometric realism alone can introduce a domain gap if the synthetic boundary statistics do not precisely match those of real lesions.

The introduction of anatomical localization priors leads to a different trade-off. While Dice does not improve substantially compared to non-localized training, precision increases and validation losses decrease (Table 2), indicating more conservative predictions. Qualitative inspection confirms fewer false positives in implausible regions such as ventricles and extra-axial spaces, directly reflecting the tissue validity constraints imposed during lesion placement (Section 3.6). This supports the interpretation that anatomical priors act as an effective inductive bias: even when overlap metrics remain similar, spatially informed simulation stabilizes training and reduces anatomically implausible errors.

The configuration that additionally incorporates intensity priors yields the highest precision among all tested setups but shows reduced performance on the synthetic subset. This behavior suggests that introducing more realistic intensity variability increases task difficulty and exposes limitations in the current intensity model, which applies relatively uniform multiplicative shifts within lesions. In practice, small lesions with low contrast were more frequently missed, particularly in deep or noisy regions, indicating that the present model does not yet capture the full heterogeneity of ADC and DWI signal changes observed clinically.

These observations motivate several targeted directions for improvement. First, lesion simulation should be driven more directly by quantitative ADC behavior rather than mask geometry alone, as ADC values encode biophysical tissue state and vary systematically within lesions. Second, explicit modeling of lesion substructure, such as a necrotic core and peri-lesional rim could reduce the tendency toward overly homogeneous synthetic contrast and improve sensitivity to small lesions. Third, stronger alignment between morphological descriptors used for statistical analysis (Section 3.8) and those used during shape optimization would reduce inconsistencies that may contribute to boundary-related domain gaps. Quantitative verification of descriptor matching, for example using Kolmogorov–Smirnov [12] tests between real and synthetic distributions, would further strengthen the validity of the shape prior.

Finally, while the present evaluation demonstrates robust generalization within ISLES, the ultimate validation of simulation-based inductive bias requires testing on completely unseen clinical cohorts acquired at different centers. Such evaluation would determine whether the observed reduction in false positives and improved precision translates into genuine robustness beyond the statistical envelope of ISLES, which remains the decisive criterion for clinical relevance.

7 Conclusion

This project evaluated whether synthetic ischemic stroke lesions, constrained by empirical statistics from ISLES-2022, can effectively support deep learning–based lesion segmentation. Several non-obvious insights emerged from the experiments.

The most robust performance on real ISLES validation data was achieved using the simplest spherical lesion model (Section 4, Table 2), which was initially unexpected given its limited visual realism. This result suggests that, under Dice-based optimization, simple and compact synthetic shapes provide a strong inductive bias for lesion localization, even when boundary realism is limited.

Increasing the geometric and intensity realism of synthetic lesions led to improved performance on synthetic data, but slightly degraded performance on real ISLES data, highlighting the presence of a domain gap. In practice, this discrepancy manifested as more fragmented segmentations, occasional false positives in anatomically complex regions, and reduced sensitivity to small, low-contrast lesions.

Accurately modelling lesions to closely match real pathological appearances is a challenging task and was not fully achieved in this project. Numerous additional factors could be incorporated into lesion modelling, and more advanced generation pipelines could be employed to better capture the complexity of real lesions. However, such refinements would require substantial additional effort and were beyond the scope of this work. Improving the realism of the synthetic lesion generation process remains a promising direction for enhancing training performance.

The main lesson of this work is that synthetic data realism must be carefully aligned with both empirical statistics and the learning objective: adding complexity without tight statistical matching can reduce generalization. In this context, controlled simplicity proved slightly more effective than partially realistic simulation.
Nevertheless, the inclusion of synthetic lesions consistently improved performance compared to the baseline model trained solely on real data, demonstrating the overall benefit of synthetic augmentation when appropriately constrained.

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